MGSC 672: Operations and Supply Chain Analytics

Individual case: Outsourcing, Near-sourcing, and Supply Chain Flexibility in the Apparel Industry

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1 Executive Summary

Since 2011 Veizenburg has existed as an online fashion. Currently it faces challenges with high demand variability, due to seasonal high-margin items referred to as limited edition collections (LECs). The LECs are often overstocked, leaving unsold units at the end of the selling season. The excess stock has storage costs and opportunity costs for other products. To further compound the issue Veizenburg can use markdown strategies to clear the remaining inventory but has concerns that aggressive markdown strategies will damage the brand.

In this case study, we will investigate how near-sourcing and out-sourcing may relieve the markdown strategy, a symptom of supply chain inflexibility and forecasting inaccuracy. The recommended approach is to adopt near-sourcing as part of its supply chain to take on variable random demand on each LEC. Near sourcing will decrease excess stock and utilize avoid aggressive markdown strategies.

2 Problem Presentation & Analysis

The Veizenburg company is focused on relatively small volumes with high prices and high margis. An example from the text demonstrates how they sold several hundred units of an LEC, for $135 with a purchase price of $65. The tradeoff of high profit is low demand predictability. The variability of demand has created an overstock problem. The issue is further compounded by the decision to offshore its production to Ukraine for cheaper products. This creates a lead time is 3 weeks for inventory. This model is dependent on forecast accuracy, which is not reliable. Instead, the company should focus on a hybrid near and out-sourcing strategy.

3 Part 1: Outsourcing business model

To best evaluate and predict the demand variability of Veizenburg we should first understand their historical data. The available data is plotted in Exhibit 1. Exibits 2 and 3 further explore the data. For exhibit 1, the line graphs show the individual sales of each LEC over time. Each LEC sale has a unique and highly variable sales curve. However, a common trait for each sale is that each has a peak sale day with many small variations thought out the cycle. For some, the peak is at the beginning of the period for others, the peak is at the end of the sale period.

The available data does allow for forecasting future demand. We will therefore forecast the demand of the next LEC using a time series model. For the simplicity of forecasting, we will use a multiplicative method which factors a de-personalized demand, trend factor and seasonality of demand. Using a Python script, we separated the seasonal and non-seasonal components of the prediction. The number of days in the LEC sale will be the trend for sales. **Predicting the sale of LEC 24 assuming a 18-day cycle results in a total sale quantity of 514 units for 21 day sale.** This quantity is within the expected amount as the average sales for a LEC is 337, with a min and max of 133 and 1,244, respectively. This also assumes that the cost of understocking is no cost of overstocking and holding inventory. If that is not the case, then this assumption may change.

Veizenburg has offshored its production, which has numerous financial benefits but some supply chain and strategic drawbacks. The LEC business has high variability and uncertainty in demand. Therefore, forecasting accurately is difficult. Veizenburg should continue to utilize the production from Ukraine to lower costs, but with a more accurate forecasting algorithm. The data can train a time series model that will more accurately predict the product required to work around this difficulty. They can enrich the sales data with external holidays or events such as markdowns to help insights and understand demand fluctuations. Veizenburg also needs to will need to optimize the production demand to reduces costs. If the cost of overstocking is minimal (ordering too much inventory and storing it) they are best positioned to overstock and pay for the unused capacity. In contrast, a high cost of overstocking would require a different approach. Given the limited information and the assumption that overstocking costs are minimal, the company should use a timer series model to predict the order quantity while adding 2 standard deviations above the average sale volume as they will likely capture above 95% of demand, including random component.

4 Part 2: Near-sourcing business model

Instead of outsourcing all the supply Veizenburg should use a combination of near sourcing and outsourcing for their LECs. The outsourced inventory should form the baseline stock, while

the local supplier forms a quick response inventory. The local suppliers have a constraint of 12 daily units, indicating the need for a careful balance between on-hand and quick response inventory. Assuming the cost of overstocking items is high, this option gives more flexibility on order quantities. For instance, instead of ordering 512 units for LEC 24, they order 500 units. If the actual demand is lower, then the company reduces inventory costs and the can meet a higher demand of up to 524 within a 2 day cycle. This strategy allows Veizenburg to reduce costs, waste and maintain a level of service. So long as the costs of overstocking are high and the 2 day level of service is acceptable, the company should infill with local suppliers to capture more demand than they otherwise could.

Compared to offshore production this strategy will cost $30 more per unit above the inventory from overseas. The cost of understocking is $70 (based on $65 cost and $135 sale) and $40 in the offshore and near sourcing respectively. Therefore, the cost of overstocking must be at least $30 for financial viability. The near sourcing reduces waste and contributes towards other company priorities and objectives. Using all resources and available supply routes reduces risk and increases flexibility. Veizenburg should adopt a hybrid supply chain as it aligns with their objectives and likely their costs.

5 Part 3: Flexibility business model

The flexibility model of the supply chain allows Veizenburg to adapt to the demand and align with strategic objectives quickly. The company can order 1,000 units in 1 week instead of the typical 3-week turnaround. The average LEC sales volume is 338 units, much lower than the 1,000 units available. The flexibility model may help overcome the risk markdown strategy and inventory overstock as this model increases the speed of the supply chain. The new-found supply chain speed decreases inventory waste by allowing greater flexibility in inventory management. The company can stagger the orders to best guage demand and reduce waste. For instance, order half of the expected quantity for an LEC and one for the remaining quantity after multiple days of sales data if the sales are going well and less if not. Thereby allowing the company to order when demand is high and not when the demand is low. This option gives high speed and flexility at a lower cost than the near-sourcing option. This option gives tighter control of the quantity of LECs and reduces waste.

Veizenburg may also benefit through secondary benefits through this strategy. The company may want to reduce waste, limit markdowns and increase inventory sell-through. A flexible and dynamic supply chain and inventory approach easily enables all of these objectives. Increasing the sell-through rate enables a lower inventory leftover, reduces waste and creates less pressure for markdowns. Instead, the company may consider flash sale vendors to dispose of excess small inventory.

6 Part 4: General Questions

Outsourcing has some financial benefits for Veizenburg, however there are some major drawbacks. A 3-week order time with low forecast accuracy is a limitation. Meanwhile near-sourcing has a quick response time, allowing Veizenburg to respond to demand more readily and reducing the pressure on accuracy, as local suppliers can fill the variable or random demand. Thereby allowing outsourcing to focus on larger, more predictable baseline demand supply and reducing excess inventory from overordering. A flexible option for the supply chain balances the cost with the supply chain speed allowing quick decisions and waste reduction via a tighter control of the supply chain. Regardless of choice, more accurate predictions in the supply chain also allows Veizenburg to deliver on its waste reduction strategies and avoid markdowns which may damage its reputation.

7 Appendix

7.1 Exhibits and Figures

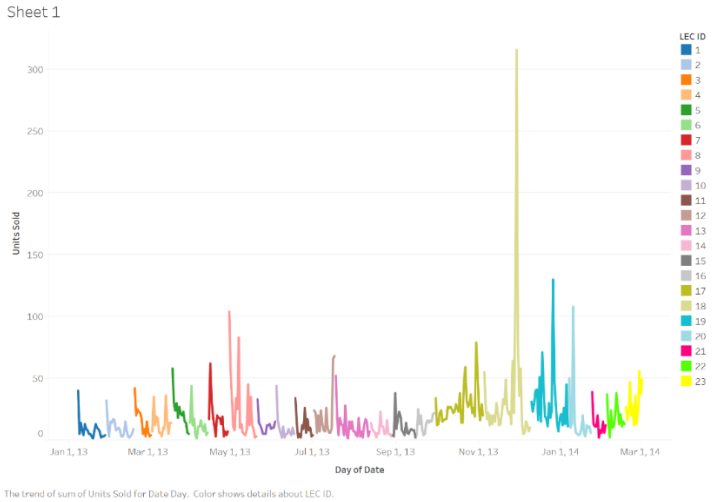


Exhibit 1: Reproduction of exhibit 2 in tableau from original case

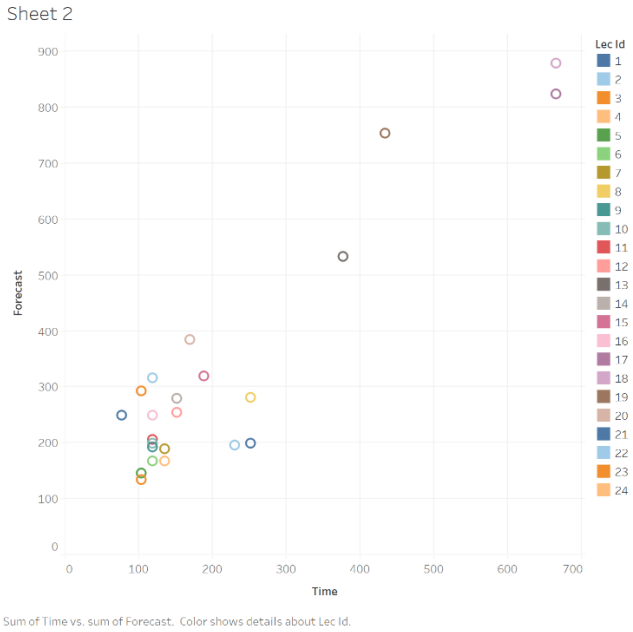


Exhibit 2: Scatter plot of Maximum time for each LEC and the number of unit sales. A linear trend emerges

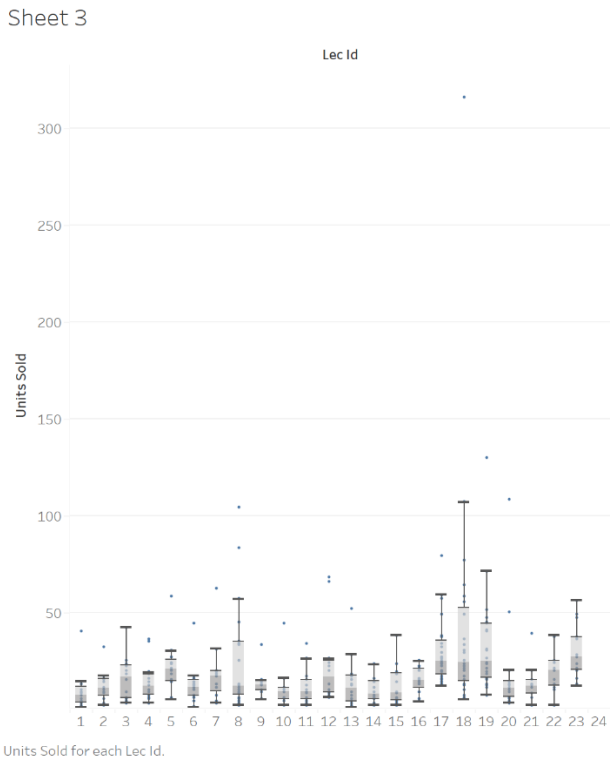


Exhibit 3: Box and whisker plot of daily unit sales for each LEC

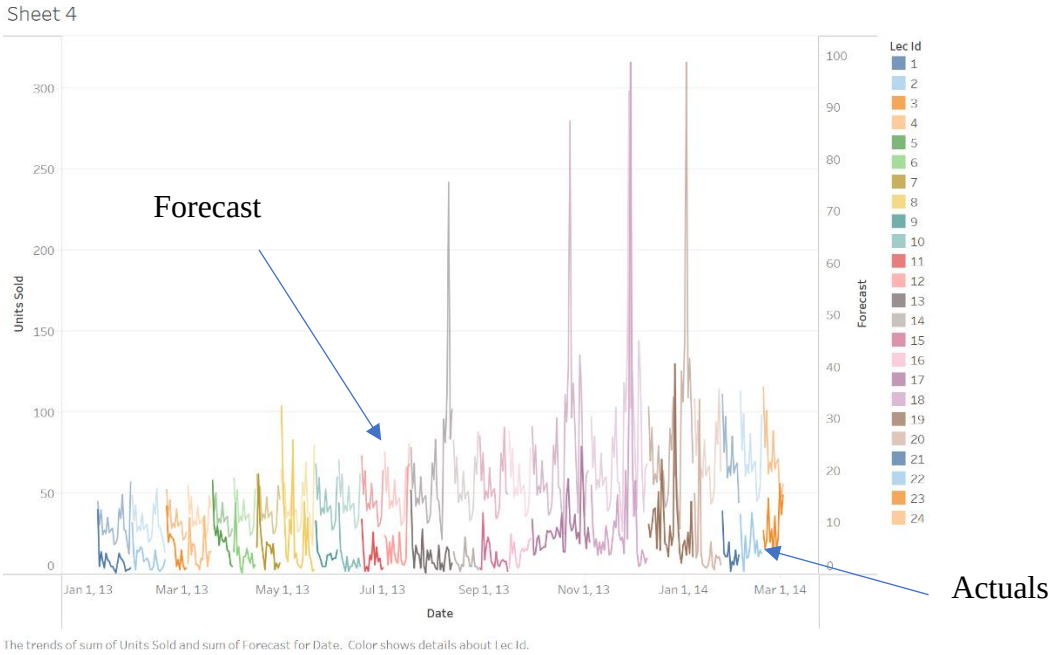


Exhibit 4: Actual sales by date and LEC with forecasted demand for each LEC based on deseasonalised demand.

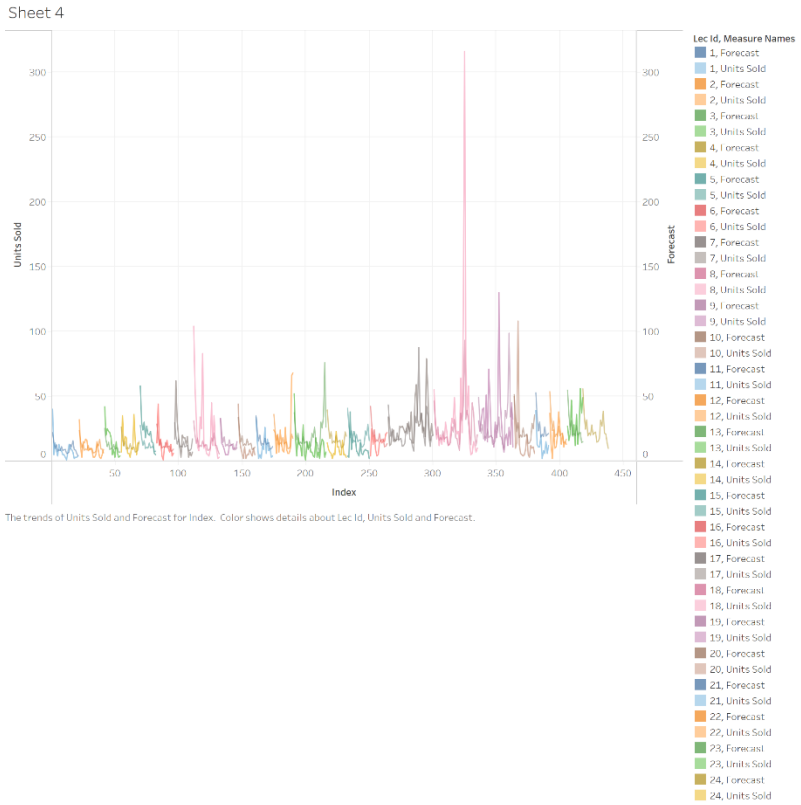


Exhibit 5: Forecast overlay with actual data. Synchronized axis

7.2 Case Background

Timm Veizenburg launched an online storefront in 2011 that sells clothing online. The Veizenburg company focused on relatively small volumes with high prices and margin items. The company has two primary types of items. First, there are the “must have” items with a stable demand, and the limited-edition collections (LEC), which are trendy and have a variable demand. An example from the text demonstrates how they sold several hundred units of a style, or LEC, which were purchased and sold for $65 and resold for $135. Therefore, the markup on these LEC products is around 2x, demonstrating a healthy profit margin. This type of split is often seen with products, where some “seasonal” products are high volume and low margin and others are low volume, high margin. The tradeoff between these two is demand predictability. “seasonal” products have more significant errors and a high volatility level than other stable products. Both help grow the brand and business in different ways.

The main challenge is that the company only sells some of its stock during each LEC season. Therefore, some units of clothing remain in inventory after the season has ended. This overstock problem is unavoidable as the company has offshored its production to Ukraine for cheaper products. However, the lead time is 3 weeks for inventory. Internally some believe they need to have a markdown strategy to clear inventory. Often markdown strategies are an effective way of reducing or eliminating inventory, as lower prices often make products more accessible and increase demand. Balancing markdown and inventory overstock is a persistent challenge that affects the industry. Leftover inventory incurs inventory storage space and the opportunity cost of additional items. Some examples of manufacturers demonstrate cost savings of $65 million by adjusting inventory stocks to appropriate levels.[1]

7.3 Limitations on strategy

The hybrid near- and outsourcing strategies increase the chances of understocking and missing potential demand. Intentionally under-ordering and infilling can work; however, the decision-making would need to be robust with the observed unit sales. The example points to an average of 40 units sold at the end of the LEC while the local supplier can only handle 12 units. Therefore 2 days' worth of infilling would require 7 days of outputting 12 units. Customers may not purchase as they feel alienated due to a different level of service than customers who order early in the season.

Second, any units of inventory that go above the inventory holding would need to be monitored and capped at specific numbers as continuously building up orders adds complications and liabilities to the supply chain unnecessarily. However, this does mean there will be an initial opportunity cost as the demand needs to be fully captured due to a lack of inventory. The costs will be lower than with the current scheme at the cost of unsatisfied demand. However, a smart marketing team would point out that the rarity of items may increase desirability and conversion to purchase. Mainly if this is communicated to consumers as a "3 units left" message. This has had a profound increase in sales throughout many industries.[2]

The last limitation is that the forecast accuracy will likely never be accurate and excess or insufficient inventory will be a persistent issue. While increasing the accuracy of the forecast will help Veizenburg better understand demand and reduce the impact of over or understocking

units, the accuracy will be less than 100%. The accuracy can be reduced by forecasting at an aggregate level. However, Veizenburg may need help doing this as their forecasts are at the product level, not the colour, size or SKU level. There are always gaps between the inventory forecast and the actual numbers that cannot be eliminated. Therefore, the company must regularly update and watch the forecast amounts to ensure optimal performance while defining business rules and deciding how to manage over and understocking situations.

8 References

(1) Banker, S. The Costs Of Excess Inventory Can Be Huge. 2016.
(2) Cialdini, R. B. *Influence, new and expanded: The psychology of persuasion*; HarperCollins, 2021.